Gradience in Innovatability: Dissociating Grammaticality and World Knowledge Effects

Introduction. *Can you flask the wine?* Even though *flask* is not lexicalized as a verb, native English speakers often assign intermediate acceptability to its verb use, suggesting that the noun *flask* may be ready to be lexicalized. However, not all nouns are equally ready. For example, *to wallet the wine* sounds worse than *to flask the wine*. What drives this gradient acceptability of innovative linguistic expressions—grammaticality or real-world knowledge, or both? Gradience in linguistic judgments has been recognized since the earliest days of generative grammar [1]. Various works offer formal characterizations (e.g., via barrier enumeration [2], feature clash [3], vector similarity [4], constraint weighting [5], activation in a Large Language Model [6]). Evidence indicates that sentence acceptability can vary continuously as a function of continuous predictors [7][8]. Some have argued that language change data favor gradient linguistic representations [9][10]. However, the explanatory value of gradient encodings in linking extra-linguistic world events to grammatical change has not been articulated. In this study, we investigated gradient acceptability in sentences with innovative denominal verbs. We hypothesized that sentence acceptability could be decomposed into a syntactic component (verbiness of denominal verbs), a semantic component (verb selectional constraints), and real-world knowledge (event-probability rating) as a foundation for understanding what drives gradient acceptability.

Methods. We conducted a sentence acceptability judgment task (S; Exp 1. N = 47) on a 7-point Likert scale (1: Completely nonsensical, 7: Completely ordinary-sounding). The critical stimuli (see Table 1) included three conditions (42 items each) manipulated based on the two factors—Lexical Status (attested denominal verb (AD) vs. unattested denominal verb (UD)) and Event Probability (Probable vs. Improbable)—1) AD-Probable, 2) UD-Probable, and 3) UD-Improbable. The filler items were adapted from [11], where the target events varied in terms of event probability—1) Probable, 2) Improbable, and 3) Impossible. Then, we conducted three additional judgment tasks on event probability/possibility (EventProb; Exp 2. N = 173) and two types of phrase acceptability (P1; Exp 3. N = 52, P2; Exp 4. N = 23) (see Table 1)—P1 probes the verbiness, a syntactic property of target denominal verbs (we identify this as the "grammaticality" of the word as a verb), while P2 probes the selectional relation between verb and its object, a semantic property. All predictors were judged on Likert scales. We used EventProb, P1, and P2 as predictors of S.

Analysis & Results. First, we conducted a correlation analysis across four measures (S, EventProb, P1, P2; item-averaged Z-scores), all of which were significantly correlated ($0.57<r\text{s}<0.86$; $p\text{s}<.001$). To assess the unique contributions of the predictor variables to the outcome variable, we conducted a hierarchical linear mixed-effects model comparison, adding fixed effects sequentially in the order of EventProb, P1, and P2 to predict sentential acceptability with random intercepts for Participant and Item from Exp 1 (see Table 2). Each fixed effect yielded a significant increase in explained variance ($p\text{s}<.001$), indicating an independent contribution of each predictor to sentential acceptability. We constructed a three-dimensional plot with EventProb, P1, and S (see Figure 2b). P2 was excluded to mitigate multicollinearity in the linear model (S ~ EventProb + P1 + P2; EventProb VIF=3.62, P1 VIF=2.48, P2 VIF=5.78). The regression plane (S ~ EventProb + P1) explained 75% of the variance (R^2=.75).

Discussion. The results provide strong evidence of gradient grammaticality of both attested and unattested denominal verbs, and continuity of the map from verb grammaticality to sentential acceptability. In addition, world knowledge, operationalized as rated event probability, showed a significant effect on sentential acceptability, suggesting that changes in a world property may induce changes in acceptability—a possibility not predicted by discrete-grammar models. Taken together, the results suggest the importance of positing a continuous space of linguistic representations [3][4][5][6] in order to address the joint roles of gradient grammaticality and world statistics in determining what can be innovated when.

References. [1] Chomsky. 1965. *Aspects of the Theory of Syntax.* MIT Press. [2] Chomsky. 1986. *Barriers.* MIT Press. [3] Fouvry. 2003. *Robust Processing for Constraint-based Grammar Formalisms*. PhD, Univ. of Essex. [4] Villata & Tabor. 2022. *Cognition*. [5] Cho et al. 2018. *CMCL 2018*. [6] Lau et al. 2017. *Cog Sci*. [7] Bresnan & Nikitina. 2008. *"The Gradience of the Dative Alternation".* [8] Wasow & Arnold. 2003. *"Post-verbal constituent ordering in English"*. [9] Bybee et al. 1994. *The Evolution of Grammar*. UCP. [10] Schmid. 2020. *The Dynamics of the Linguistic System.* OUP. [11] Hu et al. 2025. *J Mem Lang*. [12] Davies. 2008. *The Corpus of Contemporary American English (COCA)*.

Table 1. Example sentences/phrases across experiments

Exp	Example Stimuli
1 (S)	Her roommate promised to bottle/flask/wallet the wine and bring it to the picnic.
2 (EventProb)	filling a bottle/flask/wallet with wine
3 (P1)	to bottle/flask/wallet something
4 (P2)	to bottle/flask/wallet wine

Note. The critical items included three conditions: AD-Probable (e.g., *bottle*), UD-Probable (e.g., *flask*), and UD-Improbable (e.g., *wallet*). The distinction between AD and UD was made based on verb frequency in COCA [12]—words with both noun and verb frequency information were classified as AD, whereas words with only noun frequency information were classified as UD. Probable and Improbable conditions were determined based on experimenters' intuitions. For statistical analyses, we used the continuous values measured in each experiment. S: sentential acceptability, EventProb: event probability, P1: phrasal acceptability 1 ("to X something"), P2: phrasal acceptability 2 ("to X (a/an) Y").

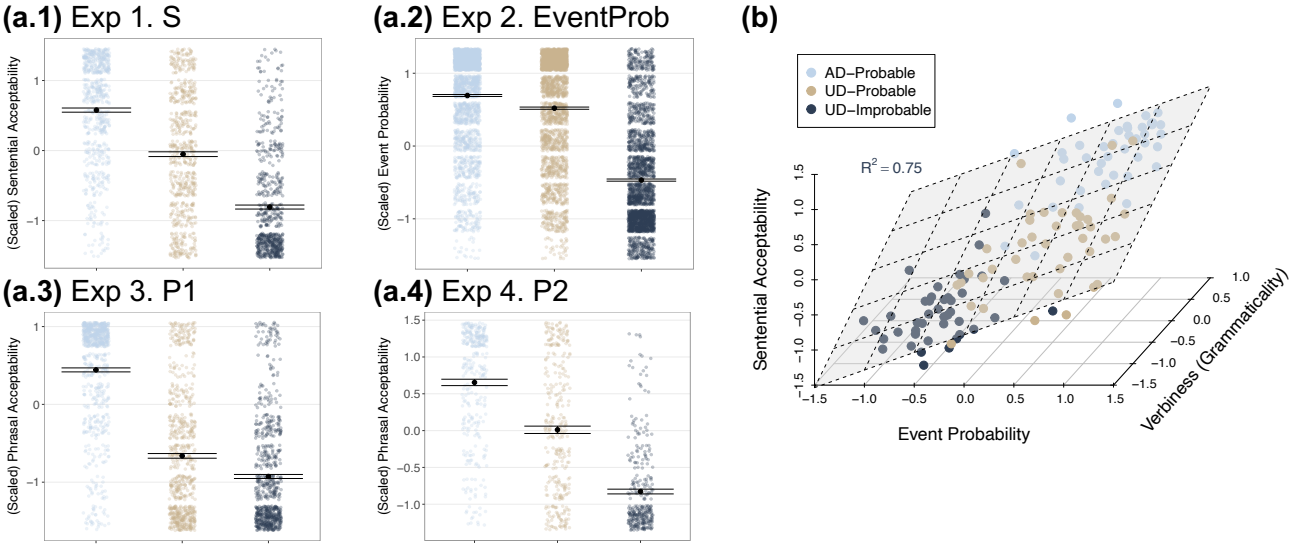

Figure 1. (a) Scaled measures (Z-scores of S, EventProb, P1, P2) across critical conditions (AD-Probable (left), UD-Probable (center), UD-Improbable (right)). Means ± SE are shown as black dots and lines. (b) 3-D plot of EventProb, P1, and S. Each point represents item-averaged Z-scores. The grey plane shows the fitted linear model (S ~ EventProb + P1), which explains 75% of the variance (EventProb VIF=1.49; P1 VIF=1.49).

Table 2. Hierarchical linear mixed-effects model comparison

	Model	R_m^2	R_c^2	ANOVA Results
M0	S ~ 1 + (1|Participant) + (1|Item)	0.00	0.57	
M1	S ~ EventProb + (1|Participant) + (1|Item)	0.29	0.57	$\chi^2(1)$=108.95, *p*<.001
M2	S ~ EventProb + P1 + (1|Participant) + (1|Item)	0.37	0.57	$\chi^2(1)$=66.76, *p*<.001
M3	S ~ EventProb + P1 + P2 + (1|Participant) + (1|Item)	0.38	0.57	$\chi^2(1)$=16.98, *p*<.001

Note. Analyses included only critical items, using participant and item information from Exp 1 (S).